

Question	Answer	Mark	Mark scheme	Additional guidance
20 (a)	graph	C2 (C1	for a translation of the graph by the vector $\begin{pmatrix} -1 \\ -3 \end{pmatrix}$ for a translation of the graph by the vector $\begin{pmatrix} -1 \\ b \end{pmatrix}$ where $b \neq -3$ or $\begin{pmatrix} a \\ -3 \end{pmatrix}$ where $a \neq -1$ or for a translation by the vector $\begin{pmatrix} -1 \\ -3 \end{pmatrix}$ of 3 or 4 critical points)	Condone graph of $y = f(-x)$ also being drawn on the grid Correct vector gets 1 mark
(b)	2, 1	B1	cao	